

1) $\lim_{x \rightarrow \infty} \frac{x^2 \cdot \cos(2x)}{x^4 + 3} = ?$

Handwritten notes: $-1 \leq \cos(2x) \leq 1$, $\frac{-x^2}{x^4(1+\frac{3}{x^4})}$, $\frac{-1}{x^2}$

a) 1 b) 0 c) Limit mevcut değildir d) 1/3 e) 2

2) $f(x) = \sqrt{2 - \sqrt{3 - \sqrt{4 - x}}}$ fonksiyonunun tanım kümesi aşağıdakilerden hangisidir?

Handwritten notes: $3 \geq \sqrt{4-x}$, $\sqrt{4-x} \leq 3$, $3 - \sqrt{4-x} \geq 0$, $4-x \geq 0$, $4 \geq x$, $4-x \leq 9$, $-5 \leq x$, $2 - \sqrt{3 - \sqrt{4-x}} \geq 0$, $2 \geq \sqrt{3 - \sqrt{4-x}}$, $4 \geq 3 - \sqrt{4-x}$

a) (3, 4) b) (-5, 4] c) [3, 4] d) [-5, 4] e) Hiçbiri

3) $\lim_{x \rightarrow 0} \frac{x^{2020} \sin 4x}{(\sin x)^{2021}} = ?$

Handwritten notes: $\frac{x \cdot x}{\sin x}$, $\frac{1}{\sin x} \cdot \sin 4x$, $\sqrt{4-x} > -1$

a) 0 b) 2 c) 4 d) 2020 e) hiçbir

4) $\lim_{x \rightarrow 1} \frac{|x-1|+x}{|x-4|} = ?$

Handwritten notes: $1^+ \frac{(x-1)+x}{4-x} \rightarrow \frac{1}{3}$, $1^- \frac{(1-x)+x}{4-x} \rightarrow \frac{1}{3}$

a) limit mevcut değildir b) 1/2 c) 1/3 d) 1/4 e) 1

5) $\lim_{x \rightarrow -\infty} \sqrt{4x^2 + 2x} + 2x = ?$

a) 0 b) ∞ c) $-\infty$ d) $\frac{1}{2}$ e) $-\frac{1}{2}$

Handwritten notes: $\frac{4x^2+2x-4x^2}{\sqrt{4x^2+2x}-2x}$, $\frac{2x}{-x(\sqrt{4+\frac{2}{x}}+2)}$, $\frac{2x}{-4x} = -\frac{1}{2}$

6) $\lim_{x \rightarrow 0} \frac{(\sin x)^{100}}{x^{99} \sin 2x}$ limitinin sonucu aşağıdakilerden hangisidir?

a) 2 b) $\frac{1}{2}$ c) 0 d) ∞

Handwritten notes: $\frac{\sin x}{\sin 2x} = \frac{1}{2}$

7) $\lim_{x \rightarrow \infty} x^2 \cdot \left(1 - \cos \frac{2}{x}\right) = ?$ a) 0 b) 1 c) 2 d) $\frac{1}{2}$

8) $\lim_{x \rightarrow 0} \frac{1 - \cos x}{\sin x \cdot \tan x} = ?$ a) 0 b) 1 c) $\frac{1}{2}$ d) Limit mevcut değil

9) $f(x) = \frac{\sqrt{12x-3} - x}{x^3 - 1}$ fonksiyonunun tanım kümesi? $(-\infty, 1) \cup [3, \infty)$

10) $\lim_{x \rightarrow \infty} \frac{x - \sqrt{1+x^2}}{x - \sqrt{x}} = ?$ a) 1 b) 2 c) 0 d) ∞ e) $-\infty$

11) $\lim_{x \rightarrow 0} \frac{1 - \sqrt{1 - \sin 2x}}{\tan x} = ?$ a) 0 b) 1 c) 2 d) ∞

12) $f(x) = \frac{\sqrt{12x+5} - 3}{x^2 + 1}$ Tanım kümesi? a) $[-1, \infty) \cup (-\infty, -4]$ b) $(-1, \infty) \cup (-\infty, -4)$ c) $(4, \infty) \cup (-\infty, 1)$ d) $[4, \infty) \cup (-\infty, 1]$

13) $f(x) = \frac{1}{\sqrt{|x| - x}}$ Tanım kümesi? a) $(0, \infty)$ b) $(-\infty, 0)$ c) $(1, \infty)$ d) $(-\infty, 1)$

14) $\lim_{x \rightarrow \infty} (\sqrt{x^2 + x} - \sqrt{x^2 - x}) = ?$ a) 0 b) 1 c) 2 d) $\frac{1}{2}$ e) -1

15) $\lim_{x \rightarrow 0} \frac{\sin x^2}{\sqrt{1+x^2}-1} = ?$ a) 0 b) 1 c) 2 d) $\frac{1}{2}$ e) -1

$\frac{\sin x^2 \cdot (\sqrt{1+x^2} + 1)}{x^2}$

16) $\lim_{x \rightarrow \infty} x^2 \cdot \sin \frac{1}{x} = ?$ a) 0 b) 1 c) 2 d) ∞

$\frac{\sin \frac{1}{x}}{\frac{1}{x}} \cdot x^2 \cdot \frac{1}{x}$ $x^2 \cdot \frac{1}{x} = x$

17) $\lim_{x \rightarrow \infty} \sqrt{x} \cdot \sin \frac{1}{x} = ?$ a) 0 b) 1 c) 2 d) ∞

$\frac{\sin \frac{1}{x}}{\frac{1}{x}} \cdot \sqrt{x} \cdot \frac{1}{x}$ $\frac{1}{\sqrt{x}}$

18) $\lim_{x \rightarrow 0^-} \frac{\sqrt{1-\cos 2x}}{\sin 2x} = ?$ a) 0 b) 1 c) $\frac{1}{\sqrt{2}}$ d) $-\frac{1}{\sqrt{2}}$

$\frac{\sqrt{1-(1-2\sin^2 x)}}{2 \sin x \cos x}$ $\frac{-\sqrt{2} \sin x}{2 \sin x \cos x}$ $\frac{-\sqrt{2}}{2 \cos x}$ $\frac{-\sqrt{2}}{2}$

19) $\lim_{x \rightarrow 1} \frac{\sqrt[3]{x}-1}{\sqrt{x}-1} = ?$ a) 0 b) 1 c) $\frac{3}{4}$ d) $\frac{4}{3}$

$\frac{\frac{1}{3} \cdot x^{-\frac{2}{3}}}{\frac{1}{2} \cdot x^{-\frac{1}{2}}}$ $\frac{\frac{1}{3}}{\frac{1}{2}} \cdot 1$

20) $\lim_{x \rightarrow 3^+} \frac{\sin \sqrt{x-3}}{\sqrt{\sin(x-3)}} = ?$ a) 0 b) 1 c) -1 d) ∞

$\frac{\sin \sqrt{x-3}}{\sqrt{x-3}} \cdot \frac{\sqrt{x-3}}{\sqrt{\sin(x-3)}}$ $\frac{\sqrt{\frac{x-3}{\sin(x-3)}}}{1} = 1.1 = 1$

21) $\lim_{x \rightarrow \infty} \left(x - \frac{x^2}{\sqrt{x^2+x}} \right) = ?$ a) 0 b) 1 c) $\frac{1}{2}$ d) ∞

$\frac{\sqrt{x^4+x^3}-x^2}{\sqrt{x^2+x}}$ $\frac{x^4+x^3-x^4}{\sqrt{x^2+x} \cdot (\sqrt{x^4+x^3}+x^2)}$

22) $\lim_{x \rightarrow 0} \frac{1-\sqrt{1+\sin x}}{1-\sqrt{1-x}} = ?$

$\frac{-\sin x}{x} \cdot \frac{(1+\sqrt{1-x})}{(1+\sqrt{1+\sin x})}$ $\frac{x^3}{x\sqrt{1+\frac{1}{x}} \cdot (x^2(\sqrt{1+\frac{1}{x}}+1))}$ $\frac{x^3}{x\sqrt{1+\frac{1}{x}} \cdot (x^2(\sqrt{1+\frac{1}{x}}+1))}$ 1.2

$-1 \cdot \frac{(2)}{2} = -1$

① $\lim_{x \rightarrow \pi} \frac{\sin 2x}{x^2 - \pi x} = ?$ a) 0 b) $\frac{2}{\pi}$ c) ∞ d) 1

$$\frac{\sin 2x}{x(x-\pi)} = \frac{2}{x-\pi}$$

$$\frac{2 \cos 2x}{2x-\pi} = \frac{2}{\pi}$$

② $\lim_{x \rightarrow -\infty} (x^2 + x \sqrt{x^2 - 1}) = ?$ a) 0 b) 2 c) $\frac{1}{2}$ d) $-\frac{1}{2}$ e) ∞

$$\frac{x^4 - x^4 + x^2}{x^2 - x \sqrt{x^2 - 1}} = \frac{x^2}{x^2 + x^2 \sqrt{1 - \frac{1}{x^2}}}$$

$$\frac{x^2}{x^2(1 + \sqrt{1 - \frac{1}{x^2}})} = \frac{1}{2}$$

③ Aşağıdaki limitlerden hangilerinin sonucu doğrudur?

I) $\lim_{x \rightarrow 1} (x-1) \cdot \sin \frac{1}{x-1} = 1$ II) $\lim_{x \rightarrow 2} \frac{\sin(x-2)}{x^2-4} = \frac{1}{4}$ III) $\lim_{x \rightarrow 0} \frac{\sin^2 2x}{x^2} = 2$ IV) $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = 0$

Yalnız II